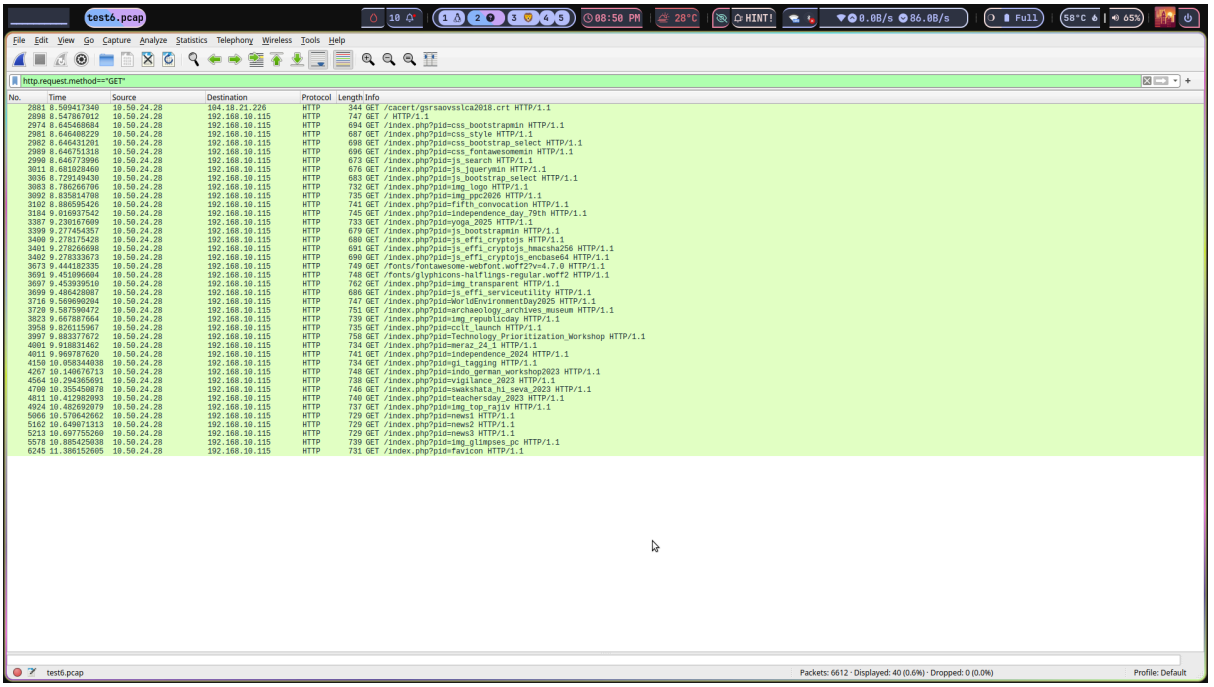
# LAB ASSIGNMENT 4
# COMPUTER NETWORK
# ID : 12341550

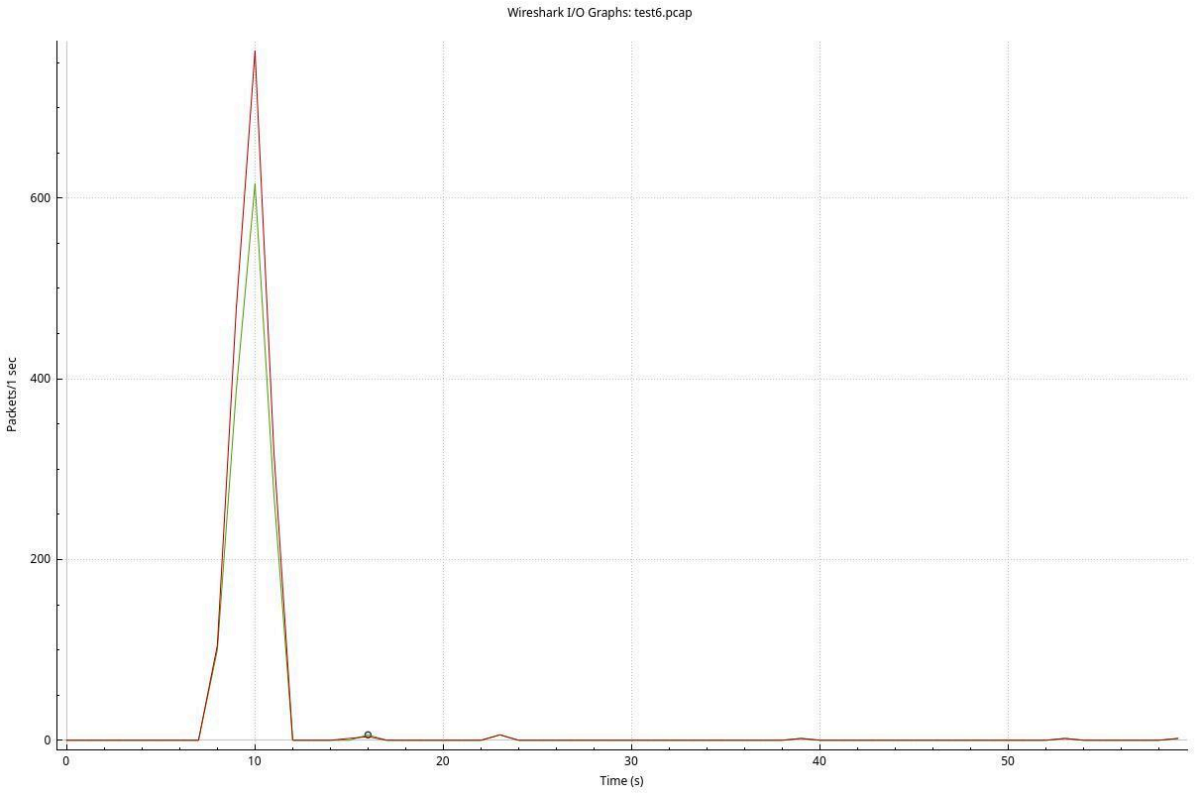
Pcap : https://drive.google.com/file/d/1X6SGRsbYkQNGtATRajRJsPl4cgi-tKd9/view?usp=sharing

## PART 1

1. **When you browse the IIT Bhilai main page (https://www.iitbhilai.ac.in) how many requests are sent (how many of the GET requests are for embedded content and how many get requests for the text)? Plot the IO graph for packets sent to iitbhilai.ac.in and packets received from iitbhilai.ac.in**




No. of GET requests packets : 40 / 6612
No. of GET requests for the text : 1 / 40




*Red line - packet sent to [iitbhilai.ac.in](iitbhilai.ac.in)*
*Green line - packet received from [iitbhilai.ac.in](iitbhilai.ac.in)*

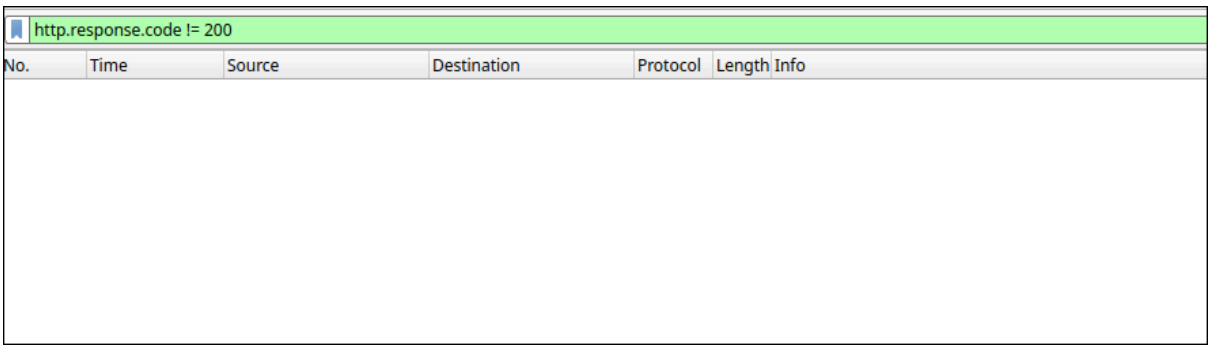
**2. For each HTTP GET request as you see above, find out the total amount of data being received in the corresponding HTTP response message.**

http.response.code==200

| o. | Time | Source | Destination | Protocol | Length | Info |
|---|---|---|---|---|---|---|
| 2888 | 8.537830720 | 104.18.21.226 | 10.50.24.28 | HTTP | 355 | HTTP/1.1 200 OK (application/x-x509-ca-ce |
| 3034 | 8.728301543 | 192.168.10.115 | 10.50.24.28 | TLSv1.2 | 4969 | HTTP/1.1 200 OK (text/css) |
| 3090 | 8.835221232 | 192.168.10.115 | 10.50.24.28 | TLSv1.2 | 16752 | HTTP/1.1 200 OK (text/css) |
| 3128 | 8.943031644 | 192.168.10.115 | 10.50.24.28 | TLSv1.2 | 7913 | HTTP/1.1 200 OK (text/css) |
| 3189 | 9.054847207 | 192.168.10.115 | 10.50.24.28 | HTTP | 1684 | HTTP/1.1 200 OK (text/css) |
| 3191 | 9.103353503 | 192.168.10.115 | 10.50.24.28 | HTTP | 948 | HTTP/1.1 200 OK (text/javascript) |
| 3397 | 9.269317162 | 192.168.10.115 | 10.50.24.28 | HTTP | 6742 | HTTP/1.1 200 OK (text/javascript) |
| 3698 | 9.484435398 | 192.168.10.115 | 10.50.24.28 | HTTP | 879 | HTTP/1.1 200 OK (text/javascript) |
| 3714 | 9.568433266 | 192.168.10.115 | 10.50.24.28 | HTTP | 1535 | HTTP/1.1 200 OK (text/javascript) |
| 3718 | 9.584757273 | 192.168.10.115 | 10.50.24.28 | HTTP | 5938 | HTTP/1.1 200 OK (text/javascript) |
| 3821 | 9.667455611 | 192.168.10.115 | 10.50.24.28 | HTTP | 4108 | HTTP/1.1 200 OK (JPEG JFIF image) |
| 3956 | 9.825615544 | 192.168.10.115 | 10.50.24.28 | TLSv1.2 | 9511 | HTTP/1.1 200 OK (image/jpeg) |
| 3995 | 9.882508595 | 192.168.10.115 | 10.50.24.28 | HTTP | 2367 | HTTP/1.1 200 OK (JPEG JFIF image) |
| 3999 | 9.917275630 | 192.168.10.115 | 10.50.24.28 | HTTP | 1677 | HTTP/1.1 200 OK (text/javascript) |
| 4009 | 9.968514318 | 192.168.10.115 | 10.50.24.28 | TLSv1.2 | 17071 | HTTP/1.1 200 OK (text/javascript) |
| 4147 | 10.057135685 | 192.168.10.115 | 10.50.24.28 | HTTP | 1178 | HTTP/1.1 200 OK (JPEG JFIF image) |
| 4265 | 10.140113750 | 192.168.10.115 | 10.50.24.28 | TLSv1.2 | 21479 | HTTP/1.1 200 OK (JPEG JFIF image) |
| 4561 | 10.293098288 | 192.168.10.115 | 10.50.24.28 | TLSv1.2 | 31155 | HTTP/1.1 200 OK (PNG) |
| 4698 | 10.354934319 | 192.168.10.115 | 10.50.24.28 | HTTP | 4076 | HTTP/1.1 200 OK (JPEG JFIF image) |
| 4809 | 10.411886662 | 192.168.10.115 | 10.50.24.28 | TLSv1.2 | 6325 | HTTP/1.1 200 OK (JPEG JFIF image) |
| 4922 | 10.482076481 | 192.168.10.115 | 10.50.24.28 | TLSv1.2 | 6317 | HTTP/1.1 200 OK (JPEG JFIF image) |
| 5576 | 10.884014346 | 192.168.10.115 | 10.50.24.28 | TLSv1.2 | 17688 | HTTP/1.1 200 OK (JPEG JFIF image) |
| 5728 | 11.039979931 | 192.168.10.115 | 10.50.24.28 | TLSv1.2 | 16077 | HTTP/1.1 200 OK (JPEG JFIF image) |
| 5976 | 11.195813265 | 192.168.10.115 | 10.50.24.28 | HTTP | 4572 | HTTP/1.1 200 OK (PNG) |
| 6196 | 11.323378497 | 192.168.10.115 | 10.50.24.28 | HTTP | 10034 | HTTP/1.1 200 OK (JPEG JFIF image) |

| Packet No. | Content Type (from Info column) | Data Received (Length in Bytes) |
|---|---|---|
| 2888 | application/x-x509-ca-cert | 355 |
| 3034 | text/css | 4969 |
| 3090 | text/css | 16752 |
| 3128 | text/css | 7913 |
| 3189 | text/css | 1684 |
| 3191 | text/javascript | 948 |
| 3397 | text/javascript | 6742 |
| 3698 | text/javascript | 879 |
| 3714 | text/javascript | 1535 |
| 3718 | text/javascript | 5938 |
| 3821 | JPEG JFIF image | 4108 |
| 3956 | image/jpeg | 9511 |
| 3995 | JPEG JFIF image | 2367 |
| 3999 | JPEG JFIF image | 1677 |
| 4009 | text/javascript | 17071 |
| 4147 | JPEG JFIF image | 1178 |
| 4265 | JPEG JFIF image | 21479 |
| 4561 | PNG | 31155 |
| 4698 | JPEG JFIF image | 4076 |
| 4809 | JPEG JFIF image | 6325 |
| 4922 | JPEG JFIF image | 6317 |
| 5576 | JPEG JFIF image | 17688 |
| 5728 | JPEG JFIF image | 16077 |
| 5976 | PNG | 4572 |
| 6196 | JPEG JFIF image | 10034 |

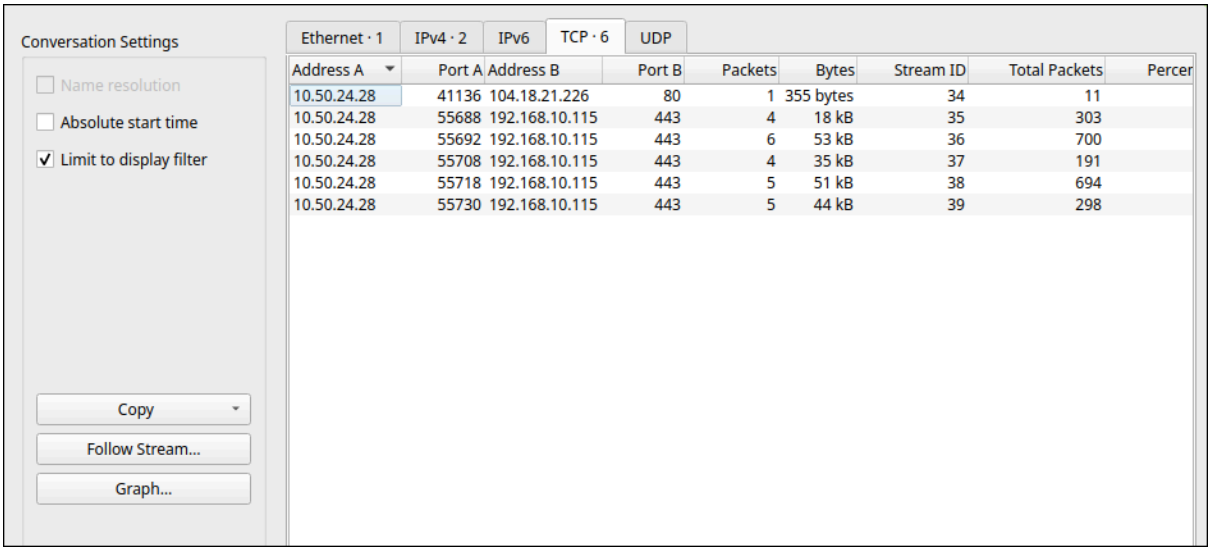
3. **How many HTTP packets were received with response codes other than 2XX? What are the different response codes received? Explain.**




Upon applying the filter http.response.code != 200 in Wireshark, no packets were returned.

Because the browser cache and history were cleared prior to the capture, the browser had no stored versions of the website's assets (images, CSS, JS). Every request required a full download from the server, leading to consistent 200 OK responses instead of redirects (301/302) or cache-validation codes (304 Not Modified).
No 4XX (Client) or 5XX (Server) errors were detected, indicating a perfectly successful page load.

4. **How many HTTP/HTTPS connections are of type persistent?**




The browser used 6 persistent connections to load the site efficiently. Instead of doing a slow "handshake" for every single file, it kept 6 "pipes" open to funnel all 60+ images and scripts through at once. We can see this in the TCP Conversations window, where only 6 rows exist despite dozens of requests, and the Connection: keep-alive header confirms the browser and server stayed "on the line" to save time.

5. **For the response to your HTTP GET request, get the image reconstructed by the hex editor.**

To reconstruct the image, I identified Packet 3821 as a JPEG JFIF image and exported its raw payload from the TCP stream. The reconstruction process involves stripping the plain-text HTTP response headers to isolate the binary image data, which is marked by the JPEG Start of Image (SOI) hex sequence FF D8 FF. Although the file command initially misidentified the raw data as a "SPARC executable" due to the un-trimmed headers, locating this magic number allows for the precise removal of metadata using a hex editor or terminal tools like dd, successfully restoring the binary payload into a viewable .jpg file.

6. **Surf a website (other than google.com) of your choice and discuss the end-to-end process of web page loading using Wireshark. How much time did it take to load the page? Find out how many connections are used to download this page. Are these connections persistent or non-persistent? How many objects have been transferred on these connections? Which object took the longest time to download?**

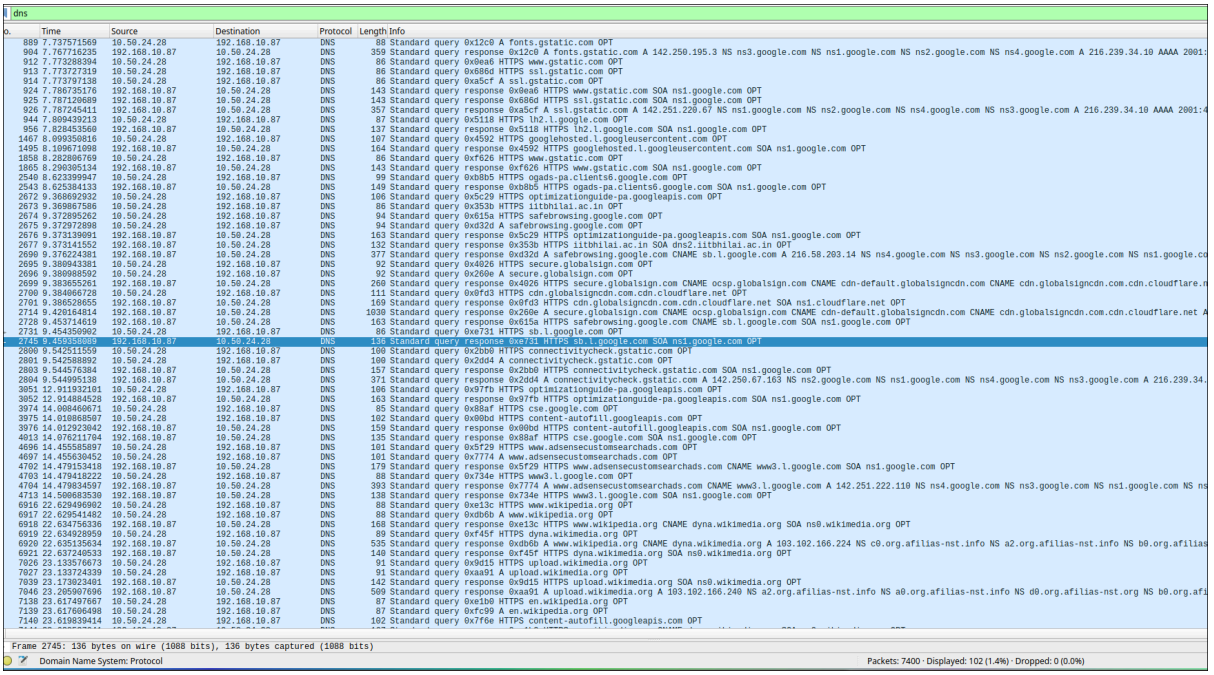
*** i didnt open any other site while capturing packets for the first time, which is essential for part 2 of the assignment so i captured it again.

Pcap : https://drive.google.com/file/d/1KoS3h-RxHh5Cn0Eh1sTenBZjDiUr3xe0/view?usp=sharing

## PART 2

1. **Along with the IIT Bhilai website, access one more website of your choice and answer the following questions.**

   **How many DNS packets have you observed in total?**




I observed a total of 102/7400 DNS packets during the capture sessions for both websites.

By examining the Destination IP of the DNS query packets, I identified the DNS server as 10.50.24.28

| A | B |
|---|---|
| Domain Name | IP Address |
| www.iitbhilai.ac.in | 192.168.10.115 |
| www.wikipedia.org | 103.102.166.224 |
| | |

2. **The root servers on the Internet are in the domain root-servers.net. You can see the list of all root servers using dig [DNS lookup utility] or any tool/command. [6 Points] Use dig to ask the root server for the address of www.iitbhilai.ac.in, without recursion. Go through the hierarchy from the root without recursion, following the referrals manually until you reach the address www.iitbhilai.ac.in. List all the name servers involved to find the IP address of www.iitbhilai.ac.in.**

```
❯ dig +norecurse @a.root-servers.net www.iitbhilai.ac.in

; <<>> DiG 9.18.39-0ubuntu0.24.04.2-Ubuntu <<>> +norecurse @a.root-servers.net www.iitbhilai.ac.in
; (2 servers found)
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 57977
;; flags: qr; QUERY: 1, ANSWER: 0, AUTHORITY: 4, ADDITIONAL: 9

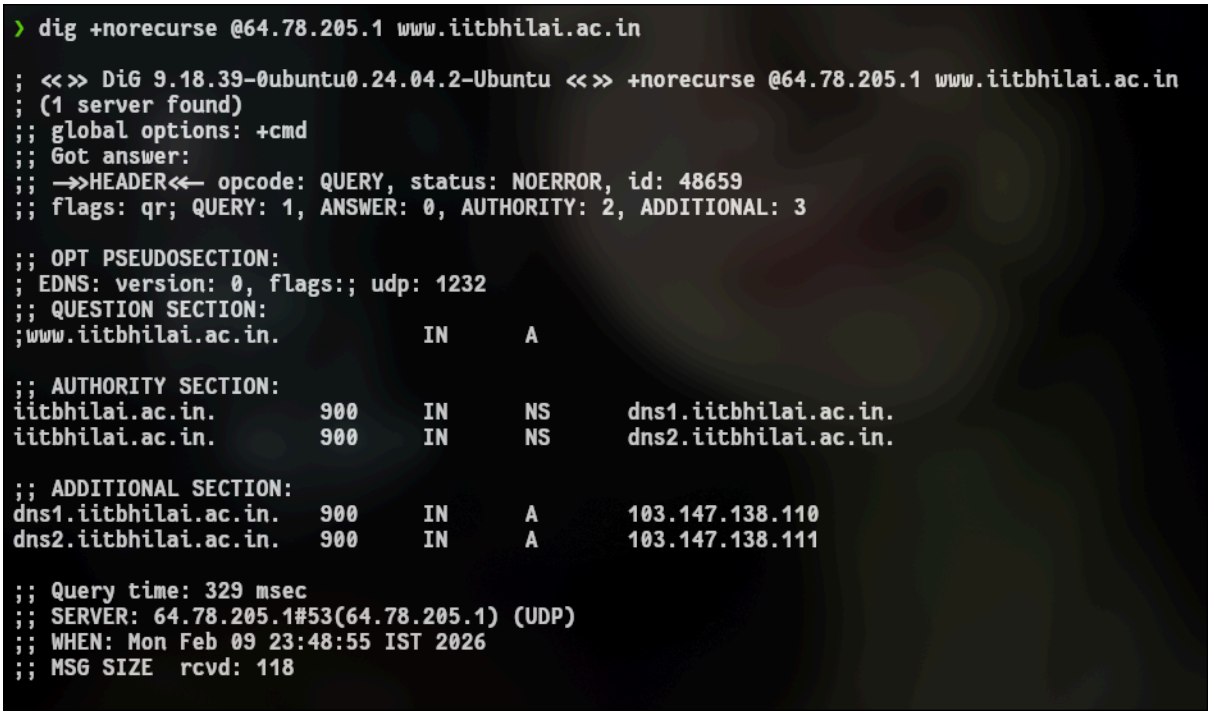
;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 4096
;; QUESTION SECTION:
;www.iitbhilai.ac.in.           IN      A

;; AUTHORITY SECTION:
in.                     172800  IN      NS      ns10.trs-dns.org.
in.                     172800  IN      NS      ns01.trs-dns.com.
in.                     172800  IN      NS      ns10.trs-dns.info.
in.                     172800  IN      NS      ns01.trs-dns.net.

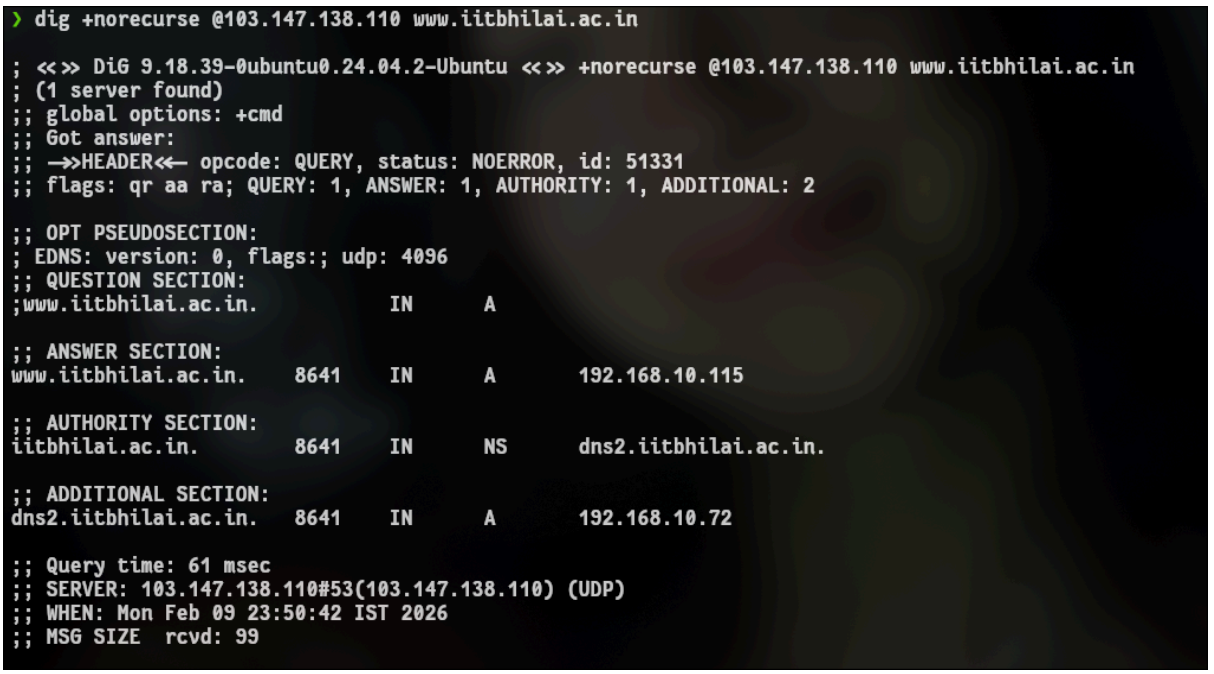
;; ADDITIONAL SECTION:
ns10.trs-dns.org.       172800  IN      A       64.78.205.1
ns10.trs-dns.org.       172800  IN      AAAA    2620:171:813:1534:8::1
ns01.trs-dns.com.       172800  IN      A       64.96.1.1
ns01.trs-dns.com.       172800  IN      AAAA    2620:57:4001::1
ns10.trs-dns.info.      172800  IN      A       64.78.204.1
ns10.trs-dns.info.      172800  IN      AAAA    2620:171:812:1534:8::1
ns01.trs-dns.net.       172800  IN      A       64.96.2.1
ns01.trs-dns.net.       172800  IN      AAAA    2620:57:4002::1

;; Query time: 289 msec
;; SERVER: 198.41.0.4#53(a.root-servers.net) (UDP)
;; WHEN: Mon Feb 09 23:46:33 IST 2026
;; MSG SIZE  rcvd: 345
```

Observation: The root server (a.root-servers.net at 198.41.0.4) does not have the specific record but redirected the request to the Top-Level Domain (TLD) servers responsible for the .in zone.




Observation: The root server (a.root-servers.net at 198.41.0.4) redirected the query to the Top-Level Domain (TLD) servers for the .in zone.




Summary of Name Servers Involved:

Root: a.root-servers.net

TLD (.in): ns10.trs-dns.org

Authoritative: dns1.iitbhilai.ac.in